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Review Article

Biological Dosimetry and Individual Susceptibility in Chronic Low-Dose Occupational Ionizing Radiation Exposure: Implications for Worker Health Surveillance

*¹Kimberly Long Holt

About Article

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About Author

¹ Health and Safety Concepts,
Environmental Safety & Health,
California, USA

Contact @ Kimberly Long Holt
longholt@slac.stanford.edu

ABSTRACT

The global number of workers who are exposed to ionizing radiation through occupation is 2.3 million in the healthcare, nuclear energy, and industrial fields. In spite of regulatory annual dose limits of 20 mSv, cellular and molecular damage are measurable at lower doses than these, with chronic low-dose exposures causing damage. Current occupational surveillance relies predominantly on physical dosimetry, which does not capture individual biological susceptibility or cumulative cellular damage. This is a synthesis of peer-reviewed sources dating from 2000 to 2025 on biological damages in occupationally exposed workers. The DNA damage indicators, such as sustained gamma-H2AX foci, micronuclei, and chromosomal anomalies, report 2-7-fold increases in health workers in the field of medicine and industrial radiography below regulatory limits. The biomarkers of oxidative stress demonstrate a sustained increase of reactive oxygen species and a 25-40 percent decrease of antioxidant defenses. The data on 308,297 nuclear workers proves that leukemia has a linear dose-response dependence and an excess risk of 0.35 per sievert. Personal risk factors such as DNA repair polymorphisms, age, and smoking also have significant effects on personal risk with the same amount of occupational exposure. The biological dosimetry techniques, such as micronuclei, comet assay, and gamma-H2AX techniques, are standardized to facilitate regular occupational health monitoring. Systematic biomarker surveillance would be helpful in improving early screening of toxic exposure and allowing protective measures that are tailored to the individual. Integrating standardized biological dosimetry into routine occupational health surveillance would enable earlier detection of radiation-induced harm and support individualized risk-based protection strategies.

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1. INTRODUCTION

Each year, about 2.3 million workers are exposed to occupational ionizing radiation in healthcare, nuclear energy, and industrial sectors worldwide (Lopes *et al.*, 2025). The group mostly exposed is medical professionals, such as radiologists, interventional cardiologists, nuclear medicine technicians, and so forth, with about 1.2 million employees in developed countries (Abbas *et al.*, 2021). Other groups of people with considerable occupational exposure include nuclear power operators, industrial radiography technicians, and non-destructive testing workers (Leuraud *et al.*, 2015, 2024). The International Labour Organization estimates that radiation occupational exposure leads to about 1,000 unnecessary deaths/radiation-induced malignancies annually. The guidelines of the International Commission on Radiological Protection introduce annual occupational limits of dose to 20 mSv (and in certain jurisdictions, 50 mSv) in a year (Tawn *et al.*, 2018). These limits are based on the linear no-threshold model, which assumes that the risk of cancer grows in proportion to the dose at all levels of exposure with no safe levels. Epidemiological evidence on occupational exposure as a risk factor for hematologic malignancies and solid cancers has been provided by historical cohort studies, particularly nuclear worker cohorts in Japan, the United Kingdom, and the United States (Leuraud *et al.*, 2024). Nevertheless, the existing regulatory systems are based mainly on a physical dosimetry based on badge measurements that are only a crude measure of exposure but do not include individual biological susceptibility and actual tissue-level harm (Baudin *et al.*, 2021). Recent studies indicate that there is a dissimilar biological pathway during chronic low-dose exposure related to acute high-dose exposure. In populations of medical workers, the evidence of observable biological responses is observed with simple annual doses of 2 to 5 mSv, which is far less than the regulated values but is enough to stimulate cellular response systems to stress (Silva-Júnior *et al.*, 2020, 2025; Zabarmawi *et al.*, 2024, 2025). This paradigm shift implies the need to adopt new evaluation models that can be used to measure dynamic biological processes instead of fixed dosimetric measurements (Młynarczyk *et al.*, 2024). This narrative review is aimed at summarizing the current scientific evidence on the cellular/molecular mechanisms of damage in the occupational worker who is occupationally exposed to ionizing radiation, the feasibility of practical biological dosimetry measures able to be used to the occupational health program, individual susceptibility factors which alter the risk and the evidence-based recommendations on the integration of biological-based monitoring systems into evidence-based worker surveillance plans.

The approach of the present study is the narrative review approach that aims at synthesizing evidence obtained in studies of human occupation, with the emphasis on peer-reviewed research studies that tested biological endpoints of chronic exposure in low doses of radiation. Research that emphasized therapeutic, accidental, or high-dose acute exposures was omitted in order to remain pertinent to the occupational environment. In comparison to earlier reviews, the work focuses on the evidence that was published within the years of 2020-25, includes the emerging biomarker standardization

efforts, and attributes a specific interest to individual factors of susceptibility that cause the biological response, which is below the established regulatory thresholds.

2. LITERATURE REVIEW

2.1. Mechanisms of damage at low doses

Ionizing radiation triggers two significant damage branches in DNA: direct damage and indirect damage caused by free radical generation. Direct effects take place where the radiation places energy into the DNA molecules, producing either single-nucleotide cuts or cross-shaft cuts (Gao *et al.*, 2020). The most severe lesions are the cases of the double-strand breaks that are naturally hard to repair correctly and lead to the creation of the translocations between the chromosomes in cases of their incorrect repair. At lower dose rate levels of occupational exposure, indirect damage pathways are predominant (Kim *et al.*, 2022). The ionizing radiation ionizes water molecules to form hydroxyl radicals, hydrogen peroxide, and other reactive oxygen species. Low dose rates enable more diffusion of ROS to occur before recombination, and this may enhance effective biological dose (Luo *et al.*, 2022). Mitochondrial dysfunction caused by radiation activates these feedback loops, which increase oxidative stress beyond the oxidative stress caused by the initial production of ROS due to radiation. Injured mitochondria enhance the release of electrons by respiratory chain complexes to generate excessive superoxide anion (Mousavikia *et al.*, 2023). This enhances oxidative stress on the mitochondrial membranes and DNA, which inhibits mitochondrial functioning. The gamma-H2AX foci indicate phosphorylation of histones at sites of DNA supranucleotide break, which can be used as the specific targets for the identification of breaks that might not be repaired or repaired slowly (Baudin *et al.*, 2021). The occupational radiation workers had higher levels of persistent foci, and medical workers had increased by 2-5 times above background (Baudin *et al.*, 2021). There are various DNA repair systems, such as base excision repair, nucleotide excision repair, and homologous recombination, that serve as a way of repairing the damage caused by radiation (Wada *et al.*, 2016). Repeated exposure to low doses over time worsens the efficiency of repair pathways by sustaining production of ROS that predisposes more lesions during the processing of repair, repair protein loss via sustained requirements, and changes in DNA repair gene expression due to epigenetics (Silva-Júnior *et al.*, 2020). The defective repair of DNA leads to genomic instability manifested in an increment in mutations, chromosomal abnormalities, and the appearance of micronuclei (Piotrowski *et al.*, 2021). Radiation-induced epigenetic alterations, such as DNA methylation changes and histone modification, remain in the occupational workers unchanged in the underlying DNA sequence (Luo *et al.*, 2022). The bystander effects are phenomena that involve the direct cell communication of non-irradiated cells adjacent to directly irradiated cells, resulting in cells being highly damaged and stressed (Abbas *et al.*, 2021).

2.2. Evidence from occupational cohorts

The scientific community of professionals, such as radiologists, interventional cardiologists, and nuclear medicine technicians,



is the group with the most significant number of occupational radiation exposures (Baudin *et al.*, 2021). These professions vary between 1 and 15 mSv annually according to specialty and facility (Silva-Júnior *et al.*, 2020). Observed studies have shown abnormally high levels of DNA damage markers among medical workers even after low levels of exposure. A multi-center analysis of 156 interventional cardiologists discovered that the levels of gamma-H2AX foci were 3.2 times greater than the controls, with the foci number being directly related to the number of years working in the field (Baudin *et al.*, 2021). Radiologists show a high frequency of micronuclei, 2.8 times higher than controls, and augmented chromosomal aberration (Silva-Júnior *et al.*, 2020). The variability of exposure in industrial radiography workers, compared to medical radiography workers, is that the workers are not subjected to consistent shielding and dose monitoring (Piotrowski *et al.*, 2021). Annual exposure is between 5-50 mSv with considerable day-to-day variation (Leuraud *et al.*, 2015). The frequencies of micronuclei found in a group of 98 industrial radiography technicians were 5-7 times higher than that of controls, and higher still (approaching levels of accident survivors) was the frequency of aberration in workers who had a dose of 10mSv every year (Piotrowski *et al.*, 2021). The occupational limitless decades-long chronic exposure occurs during the work of nuclear power plant workers and employees who carry out reactor decommissioning (Kim *et al.*, 2022). In the INWORKS partnership, the researchers studied health outcomes among 308,297 workers in the nuclear industry in France, the United Kingdom, and the United States (Leuraud *et al.*, 2024). Revised results indicated that there were significant risks of leukemia at occupational exposures, with an average of 19.4 mSv cumulative dose. In this occupational range, the dose-response relationship was still approximately linear, with the relative risk of leukemia being 0.35 per sievert, including latency periods (Leuraud *et al.*, 2024). Persistent gamma-H2AX foci, high levels of biomarkers of oxidative stress, such as 1.8-fold malondialdehyde increase, and lymphocyte chromosomal aberrations are found in biological dosimetry research of nuclear workers (Zabarmawi *et al.*, 2025). The occupational giant groups represent the best to evaluate the relationship between occupational radiation exposure and determinable cancer-related outcomes and correlate with occupational findings of adequate dosimetry (Leuraud *et al.*, 2015, 2024).

2.3. Dose-response relationships in real-world settings

The linear no-threshold model has the expectation that the cancer risks proportionally rise with the dose, and there is no safe dose (Wada *et al.*, 2016). There is empirical evidence indicating that strict linearity is mixed in the occupational range of less than 20 mSv of annual exposure. Specific biomarkers, such as unresolved gamma-H2AX foci and the micronuclei frequency, show dose-dependent increases with a dose response that appears to be linear (Baudin *et al.*, 2021). The INWORKS study with the nuclear worker cohort showed that the risk of leukemia was approximately linear with the occupational exposures of 19.4mSv, with a mean of 19.4mSv, which is corroborated by the LNT assumptions in the occupational range (Leuraud *et al.*, 2024). The biomarker increases of medical worker studies at 2

to 5 mSv of dose indicate annual dose that are higher than the predictions of the LNT dose response relationship which may be because it is more sensitive to low dose rates or because individual characteristics of susceptibility are not accounted by models of LNT at the population level (Abbas *et al.*, 2021; Zabarmawi *et al.*, 2024).

2.4. Individual susceptibility factors

Genetic polymorphisms in the genes that encode repair proteins differ significantly between individuals regarding their DNA repair capacity (Baudin *et al.*, 2021). The ATM gene forms ataxia-telangiectasia mutated protein, which is a critical sensor of DNA double-strand breaks. Carriers of variations related to low ATM expression exhibit high levels of baseline gamma-H2AX foci and amplified resistance to occupational radiation exposure (Baudin *et al.*, 2021). Homologous recombination repair capacity is influenced by BRCA1 and BRCA2 polymorphisms. The XPD gene encodes a nucleotide excision repair protein with a number of specific polymorphisms that are linked to low repair efficiency (Silva-Júnior *et al.*, 2020). Medical workers with low-activity DNA repair polymorphism show an increase of the biomarkers 1.5 to 3-fold relative to those with optimal repair genotypes at the same cumulative exposures (Baudin *et al.*, 2021). The radiation sensitivity is also dependent on age in a number of ways. The decrease in the expression of DNA repair genes with age diminishes the mending ability in aged workers (Kim *et al.*, 2022). The shortening of telomeres reduces the replicative life span, and cells become susceptible to senescence when exposed to radiation. Workers aged 55 and above exhibit 1.5- 2-fold higher gamma-H2AX foci, but the difference is not observed in workers aged below 35 with the same exposure level (Mousaverik *et al.*, 2023). Smoking results in chronic oxidative stress and the destruction of antioxidant defenses and contributes considerably to the amount of oxidative impact of radiation on occupational radiation exposure (Zabarmawi *et al.*, 2025). Smokers also show an increased level of oxidative stress biomarkers 2-3 times higher than non-smoking co-workers (Gao *et al.*, 2020). The sensitivity to radiation is impacted by body mass index due to metabolic repercussions such as mitochondrial dysfunction and oxidative stress (Luo *et al.*, 2022). Diabetes, cardiovascular disease, and pulmonary syndrome significantly increase occupational radiation sensitivity (Mousaverikia *et al.*, 2023). Diabetes also compromises the functioning of the mitochondria and the activity of the antioxidant prior to radiation exposures, which produces synergies with occupational radiation (Zabarmawi *et al.*, 2024). Micronuclei frequencies and frequencies of chromosomal aberrations in diabetic workers are considerably high. Radiation sensitivity is also similarly constant with cardiovascular disease as a manifestation of systemic endothelial dysfunction (El-benhawy *et al.*, 2022). Although similar results have been observed, there are numerous limitations associated with current biomarker studies. In many occupational cohorts, the sample sizes used are reasonably small and thus do not offer much statistical power and generalizability. The prevailing designs include cross-sectional designs, limiting causation analysis, and longitudinal risk evaluation. Also, not all studies use the same laboratory



protocols and biomarker cut-offs, which creates a challenge in inter-study comparison and the supposed need to employ standardized methodologies and bigger multicenter studies.

3. METHODOLOGY

This narrative review analyzed peer-reviewed articles that were published between 2000 and 2025 in relation to occupational radiation exposure, biological dosimetry, and occupational health outcomes. The literature search initially identified 312 records across PubMed, Scopus, and Google Scholar. After removal of duplicates and screening based on titles and abstracts, 41 articles underwent full-text review. Eighteen studies met the final inclusion criteria and formed the core evidence base of this review. The databases searched were PubMed, Scopus, and Google Scholar, despite having search terms such as: occupational radiation exposure, biological dosimetry, DNA damage biomarkers, occupational health surveillance, medical worker radiation, nuclear worker groups, and low dose radiation effects. Inclusion criteria were human occupational studies on workers exposed to ionizing radiation in medical, nuclear, or industrial environments, where measurements of biological endpoints (such as DNA damage, oxidative stress, chromosomal aberrations, or health outcome) were assessed (Abbas *et al.*, 2021; Baudin *et al.*, 2021; Leuraud *et al.*, 2024). Research was enrolled irrespective of the degree of exposure or the occupational industry. The reduction criteria were that patients with radiotherapy, survivors of radiation accidents, studies performed on animals or in vitro that lacked occupational interest, and those studies that studied high-dose therapeutic and acute exposures were excluded. All of the included research encompassed workers experiencing chronic occupational exposure at levels below acute lethal exposure (Lopes *et al.*, 2025). The main evidence base consisted of 18 published studies, which were supported by the relevant literature on the mechanisms and a review of the literature (El-benhawy *et al.*, 2022; Gao *et al.*, 2020; Kim *et al.*, 2022; Luo *et al.*, 2022; Mansouri *et al.*, 2024; Mousavikia *et al.*, 2023; Młynarczyk *et al.*, 2024; Piotrowski *et al.*, 2021; Silva- Occupational cohort data included subjects who had exposures of 1 to 50 mSv/year, with a total of 52 to 308,297 subjects. The measured biological endpoints were gamma-H2AX foci, frequency of micronuclei after cytokinesis inhibition, strand breaks of the

DNA, chromosomal aberrations by cytogenetics, biomarkers of oxidative stress, interleukins, and tumor necrosis factors (Baudin *et al.*, 2021). Results were synthesized in narrative format, and the evidence was grouped by mechanism, occupational groups, and practical applications in health surveillance. Results of data summarization comprised the compilation of three tables encompassing the outcome of biomarker analysis in work with occupational cohorts and outputs of the effect, along with their reference. PubMed and Scopus were selected for their comprehensive coverage of biomedical and occupational health literature, while Google Scholar was included to capture interdisciplinary and emerging studies. Grey literature was not systematically included due to variability in methodological rigor and limited availability of validated biological endpoint data.

4. RESULTS AND DISCUSSION

4.1. DNA damage biomarker results

Thirteen work groups with 3,472 exposed workers and 456 control workers reported consistent and elevated DNA damage biomarkers (Baudin *et al.*, 2021; Silva-Junior *et al.*, 2020; Kim *et al.*, 2022). The measure of gamma-H2AX foci indicated a 2 to 4.1-fold increment in workers who were occupationally exposed, relative to controls (Baudin *et al.*, 2021). Interventional cardiologists and radiologists had shown 3.2-fold and 2.8-fold foci increases, respectively (Baudin *et al.*, 2021; Silva-Júnior *et al.*, 2020). The workers in industrial radiography demonstrated the most significant levels with 5 to 7-fold micronuclei frequency changes (Piotrowski *et al.*, 2021). The results of the Comet assay revealed that there were higher DNA strand breaks among the chronically exposed workers, and the tail moment value was higher in 78 percent of the exposed cohort (Zabarmawi *et al.*, 2024). The aberrations with chromosomes remained years following exposure, and stable translocations and dicentrics increased dose-dependently following years of cumulative occupation exposure (Tawn *et al.*, 2018; Młynarczyk *et al.*, 2024). The most extensive epidemiological data set was the INWORKS nuclear worker group (308, 297 workers); there were linear dose-response effects, with excess risk relative to 0.35 per sievert of leukemia mortality and 0.27 per sievert of lymphoma mortality, adjusting for periods of latency of 5 to 60 years (Leuraud *et al.*, 2024; Leuraud *et al.*, 2015).

Table 1. DNA Damage biomarkers in occupational radiation workers

Occupational Group	N	Biomarker	Finding	Reference
Medical Workers (Interventional)	156	gamma-H2AX foci	3.2-fold elevation vs controls	Baudin <i>et al.</i> (2021)
Radiologists	243	Micronuclei frequency	2.8-fold vs controls	Silva-Junior <i>et al.</i> (2020)
Nuclear Medicine Tech	87	Chromosomal aberrations	4.1-fold elevation	Kim <i>et al.</i> (2022)
Industrial Radiography	98	Micronuclei + Comet assay	5-7 fold micronuclei	Piotrowski <i>et al.</i> (2021)
Healthcare Workers (Mixed)	312	gamma-H2AX foci persistence	Elevated at <5 mSv annual	Abbas <i>et al.</i> (2021)
Hospital Workers	428	Comet assay tail moment	Increased DNA strand breaks	Zabarmawi <i>et al.</i> (2024)

Radiation Workers (Saudi)	245	Oxidative DNA damage (8-OHdG)	2.4-fold elevation	Zabarmawi <i>et al.</i> (2025)
Hospital Radiation Staff	167	Micronuclei + Abnormal mitosis	3.2-fold micronuclei	Mousavikia <i>et al.</i> (2023)
Nuclear Power Workers	308297	Leukemia risk	ERR 0.35 per Sv	Leuraud <i>et al.</i> (2024)
Tritium Exposed Workers	52	Dicentric aberrations	Dose-dependent increase	Tawn <i>et al.</i> (2018)
Low-dose Exposure Study	891	Gene expression profiles	Altered stress response pathways	Luo <i>et al.</i> (2022)
Hospital Workers	156	Antioxidant status + cytogenetic	Reduced SOD/CAT, elevated damage	Gao <i>et al.</i> (2020)
Healthcare Professionals	Meta-analysis	DNA damage synthesis	Consistent elevation across studies	Baudin <i>et al.</i> (2021)

4.2. Oxidative stress biomarker results

Nine occupational cohorts with 2,534 exposed workers showed a stable increase in the oxidative stress biomarkers (Zabarmaw *et al.*, 2025; Mansouri *et al.*, 2024; Gao *et al.*, 2020). The level of malondialdehyde was 1.8 times higher in Saudi health workers, and the products of lipid peroxidation among radiography workers in hospitals were higher (Zabarmawi *et al.*, 2025; Kim *et al.*, 2022). Eight-hydroxy-2-deoxyguanosine, which is a similar indicator of oxidative damage to DNA, was found to be 2.1 times higher than in controls in hospital workers (Zabarmawi *et al.*, 2024). The level of enzyme antioxidant activities decreased significantly, and the superoxide dismutase activity was downgraded by 35-40 percent and catalase activity by 28-32 percent in the hospital radiation employees and hospital workers, respectively (Mousavikia *et al.*, 2023; Gao *et al.*, 2020). The

level of glutathione, an essential body antioxidant, declined 25 to 30 percent in low-level workers (Abbas *et al.*, 2021). The meta-analytic review of 12 articles confirmed that oxidative stress indices were commonly raised in various occupational groups of people (Mansouri *et al.*, 2024). A low-dose exposure study found four similarities, with gene expression profiling revealing increased pathways of oxidative stress response and reactive oxygen species-regulated changes in gene expression of 891 workers (Luo *et al.*, 2022). Diametrically dose-related deterioration was also observed in the total antioxidant status of medical workers in relation to accruing occupational exposure (Silva-Júnior *et al.*, 2020). Such results suggest that persistent occupational radiation causes excessive self-saturation of antioxidant cellular systems and persistent oxidative stress, as well as the loss of cellular homeostasis (Zabarmaawi *et al.*, 2024, 2025).

Table 2. Oxidative stress biomarkers in occupational radiation workers

Study Population	N	Biomarker	Result	Reference
Healthcare Workers (Saudi)	245	MDA (Malondialdehyde)	1.8-fold elevation	Zabarmawi <i>et al.</i> (2025)
Hospital Workers	428	8-OHdG (oxidative DNA damage)	2.1-fold vs controls	Zabarmawi <i>et al.</i> (2024)
Hospital Radiation Staff	167	SOD activity (antioxidant enzyme)	Reduced 35-40%	Mousavikia <i>et al.</i> (2023)
Hospital Workers	156	CAT activity (catalase)	Reduced 28-32%	Gao <i>et al.</i> (2020)
Medical Workers	243	Total antioxidant status	Dose-dependent decline	Silva-Junior <i>et al.</i> (2020)
Hospital Radiography	87	Lipid peroxidation products	Significantly elevated	Kim <i>et al.</i> (2022)
Healthcare Professionals	Meta-analysis	Oxidative stress indices	Consistent elevation across 12 studies	Mansouri <i>et al.</i> (2024)
Occupational Cohort	891	ROS-related gene expression	Enhanced oxidative stress pathways	Luo <i>et al.</i> (2022)
Low-dose Workers	312	Glutathione depletion	Reduced 25-30%	Abbas <i>et al.</i> (2021)

4.3. Inflammatory marker results

Seven groups of occupational cohorts consisting of 2,181 exposed workers have shown the current systemic inflammatory energy rise (Zabarmawi *et al.*, 2025; Mansouri *et al.*, 2024; Silva-Júnior *et al.*, 2020). Interleukin-6 was increased

2.3 times among Saudi healthcare workers, which shows increased pro-inflammatory signaling (Zabarmawi *et al.*, 2025). Hospital workers demonstrated an increase of tumor necrosis factor-alpha 1.9-fold (Zabarmaawi *et al.*, 2024). The medical workers have recorded lymphocyte count abnormalities such as

decreased total lymphocyte counts, as well as changes in T-cell subsets ratios (Silva-Júnior *et al.*, 2020). The pro-inflammatory cytokine scales showed dose-dependent increases in hospital radiation personnel about a cumulative occupational dose (Mousaverikia *et al.*, 2023). Transcriptomics of systemic inflammation was associated with signature gene products of gene signatures of inflammatory processes. It increased the pathways of the inflammatory response in 891 employees of low-dose exposure cohorts (Luo *et al.*, 2022). The evidence of T-cell imbalance and low immune cell competence showed possible immunosuppression in the population that is exposed chronically (Abbas *et al.*, 2021). C-reactive protein moderately increased when statistically combined among studies, and meta-analysis of healthcare professionals presented modest increases but no statistical significance in all the cohorts (Mansouri *et al.*, 2024). These inflammatory marker increases indicate that chronic occupational radiation exposure initiates long-term systemic inflammatory responses, which add to the accumulating health risk processes other than direct DNA damage mechanisms (Zabarmawi *et al.*, 2024, 2025).

Table 3. Inflammatory markers in occupational radiation workers

Population	N	Marker	Effect	Reference
Healthcare Workers (Saudi)	245	IL-6 (Interleukin-6)	Elevated 2.3-fold	Zabarmawi <i>et al.</i> (2025)
Hospital Workers	428	TNF-alpha (Tumor necrosis factor)	1.9-fold elevation	Zabarmawi <i>et al.</i> (2024)
Healthcare Professionals	Meta-analysis	CRP (C-reactive protein)	Modest elevation, inconsistent	Mansouri <i>et al.</i> (2024)
Medical Workers	243	Lymphocyte count abnormalities	Reduced total, altered subsets	Silva-Junior <i>et al.</i> (2020)
Hospital Radiation Staff	167	Pro-inflammatory cytokine panel	Dose-dependent elevation	Mousavikia <i>et al.</i> (2023)
Occupational Cohort	891	Systemic inflammation gene signature	Enhanced inflammatory response	Luo <i>et al.</i> (2022)
Low-dose Exposure	312	Immune cell dysregulation	T-cell imbalance, reduced immunity	Abbas <i>et al.</i> 2021

4.4. Discussion of findings

Occupational exposure to chronic low-dose radiation has been proven to cause measurable cellular and molecular damage despite exposure levels below current regulatory limits of 20 mSv/year, pooled with evidence of 18 occupational cohorts (313,000 or more workers) proving that occupational exposure to low-dose radiation pertains to the effect, even below 20 mSv/year thresholds (Baudin *et al.*, 2021; Leuraud *et al.*, 2024). The biomarkers of DNA damage exhibit stable 2 to 7-fold increases in various occupational categories, with medical practitioners being those who present baseline increases with steady yearly exposures of 2 to 5 mSv (Abbas *et al.*, 2021; Zabarmaawi *et al.*, 2024, 2025). Clinically and epidemiologically, these elevations are not subclinical laboratory observations. The threefold and more significant rise in the frequency of micronuclei is indicative of chromosomal instability, which has been consistently linked with the augmented cancer risk in long-term prospective cohort studies. The sustained presence of foci of gamma-H2AX suggests unresolved DNA double-strand fissions, which are a hallmark initiating event in carcinogenesis, especially the maintenance, which persists over years of employment. Such changes in biomarkers are thus earlier biological markers of augmented vulnerability to illness instead of a sign of adaptation. The most significant changes shown by industrial radiography workers with the most considerable exposure variability indicate that dosimetry might not adequately predict effective biological dose (Piotrowski *et al.*, 2021). INWORKS nuclear

worker group offers the final epidemiological support with the best dose-response relationships between leukemia mortality and lymphoma population and excess relative risk of 0.35 per sievert and proves the linear no-threshold hypothesis in the workplace (Leuraud *et al.*, 2024; Leuraud *et al.*, 2015). Individual susceptibility phenomena evenly alter personal risk. Biomarker elevations also need to be interpreted, taking into consideration possible confounding factors. Examples of lifestyle factors that lead to oxidative stress and inflammatory biomarker expression include smoking, being overweight, and a lack of physical exercise. Occupational co-exposures, which can be chemical agents, shift work, and psychosocial stress, may also change biological responses to radiation. Even though some of them also controlled the age and smoking status, residual confounding is likely and supports the application of longitudinal design and personalized surveillance strategies. The workers with low-activity DNA repair polymorphisms, elderly age (more than 55), smoking status, and the presence of diabetes are the groups with 1.5 to 3-fold higher biomarker increase under the same occupational exposures (Baudin *et al.*, 2021; Mousavikia *et al.*, 2023; Zabarmawi *et al.*, 2025; Gao *et al.*, 2020). The example of biological dosimetry is sufficiently standardized by the collaborations on the international level and can be used as a tool of routine occupational health surveillance (Baudin *et al.*, 2021; Zabarmaawi *et al.*, 2024). Measurement of micronuclei, comet assay, and gamma-H2AX detection can offer a complement to physical dosimetry by measuring expression of variation in biological responses per individual and dose-rate impact not evident through badge-

based measurements (Lopes *et al.*, 2025). Systemic biomarker monitoring in high-exposure jobs would improve the earlier identification of the harmful exposures and allow customized protection measures according to personal sustainability, as well as offer objective health monitoring to complement physical dosimetry (El-benhawy *et al.*, 2022; Luo *et al.*, 2022). Its practical implementation will need to invest in laboratory capacity, training of occupational health professionals, and regulatory framework development that will set standardized biomarker protocols across jurisdictions (Leuraud *et al.*, 2015; Lopes *et al.*, 2025).

5. CONCLUSION

This synthesis of 18 occupational cohorts shows beyond any reasonable doubt that chronic low-dose occupational ionizing radiation exposure causes quantifiable cellular and molecular damage even below regulatory levels (Baudin *et al.*, 2021; Leuraud *et al.*, 2024; Lopes *et al.*, 2025). Persistent gamma-H2AX foci, micronuclei, and chromosomal aberrations are consistent DNA damage markers, which reveal 2- to 7-fold increases, even in the case of 2 to 5 mSv of annual exposure, under the 20 mSv legal dose (Silva-Junior *et al.*, 2020; Piotrowski *et al.*, 2021; Kim *et al.*, 2022). Persistent increases of reactive oxygen species and 25 to 40 per cent loss of antioxidant enzyme activity are biomarkers of oxidative stress that show chronic metabolic stress (Zabarmawi *et al.*, 2024, 2025; Mousavikia *et al.*, 2023; Mansouri *et al.*, 2024). Raising of inflammatory markers, such as a 2.3-fold increase in interleukin-6, indicates activation of immune response pathways on a systemic basis (Zabarmawi *et al.*, 2025). The findings of 308,297 nuclear workers prove that leukemia has linear dose-response relationships alongside an excess relative risk of 0.35 per sievert, which proves the validity of the linear no-threshold model of occupational radiation protection (Leuraud *et al.*, 2024).

However, the evidence base is dominated by cross-sectional occupational studies, which limits causal inference and the ability to assess long-term disease progression.

Personal factors such as DNA repair polymorphisms, age (above 55 years), smoking, and baseline diabetes are also significantly altering individual risk at the same exposures, making the need to introduce individual health surveillance strategies (Baudin *et al.*, 2021; Mousavikia *et al.*, 2023; Wada *et al.*, 2016). Biological dosimetry assays such as micronuclei, comet assay, and gamma-H2AX have been made standard to foster the use of routine implementation of occupational health surveillance (Baudin *et al.*, 2021; Zabarmawi *et al.*, 2024). Badge-based physical dosimetry does not give adequate information on the fundamental biological implications of occupational exposure and relative vulnerability of individuals (Abbas *et al.*, 2021; Luo *et al.*, 2022). A combination of the biological and physical dosimetry would offer better exposure evaluation and health surveillance than both separately (Gao *et al.*, 2020; El-benhawy *et al.*, 2022). A systemic biomarker monitoring in high-exposure jobs would improve early warning of the adverse exposures, personalised protective measures according to personal susceptibility factors, and objective health surveillance (Kim *et al.*, 2022; Tawn *et al.*, 2018). Biological dosimetry would have to be implemented as a standard practice in the field of

occupational radiation protection, and it would thus necessitate a cost in laboratory capacity, training of occupational health professionals, and development of regulatory frameworks (Leuraud *et al.*, 2015; Lopes *et al.*, 2025). Further investigations must develop standardized frequencies of screening biomarker thresholds with respect to exposure level and individual susceptibility, apply genetic susceptibility testing as part of individual surveillance measures, and standardize biological dosimetry thresholds across national regulatory borders (Młynarczyk *et al.*, 2024).

Future research should prioritize large-scale longitudinal cohort studies integrating repeated biological dosimetry with physical dose records, as well as investigations into genetic and epigenetic susceptibility markers to refine individualized occupational radiation risk assessment. It is an evidence-based curriculum that would improve the safeguarding of millions of workers who were subjected to persistent occupational radiation in the health, nuclear energy, and industrial fields (Lopes *et al.*, 2025).

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